Set-1

MORADABAD INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Sessional Test – 1

Course: B.Tech. **Semester: 3rd**

Session: 2025-2026 **Section: A, B, C, D**

***Subject* Name: DSTL** **Subject Code: BCS303**

Max. Marks: 20 **Time: 60 min**

Q. No.	1	2	3	4	5	6
CO	1	1	2	1	2	3
Bloom's Cognitive Level	K3	K3	K3	K3	K3	K3

Section(A)

Attempt all questions.

Q1. Show that for any two sets A and B: $(A \cup B)' = A' \cap B'$ using the method of mutual subsets. **(2 marks)**

Q2. If R is a relation on the set of integers defined by $R = \{(x, y): (x \cong y) mod\ m\}$, where $\cong$ denotes congruence modulo m, then prove that R is an equivalence relation. **(3 marks)**

Q3. Let A and B be two sets. If $f: A \to B$ is one-to-one and onto, prove that $f^{-1}: B \to A$ is also one-to-one and onto. **(3 marks)**

Section(B)

Attempt all questions. Each question carries 4 marks. **(3*4=12)**

Q4. Show that D_{64}, denoting the set of divisors of 64 ordered by divisibility, forms a lattice $(D_{64}, /)$.

Q5(i). Solve the following Boolean function using a K-map:

$$F(A, B, C, D) = \sum (0,1,2,5,7,8,9,10,13,15).$$

(ii). Also, express the following Boolean expression in conjunctive normal form:

$$(x + y + z)(xy + x' z)'$$

Q6(i). Show that $\left((p \vee q) \wedge (p' \wedge (q' \vee r'))'\right) \vee (p' \wedge q') \vee (p' \vee r)$ is a tautology.

(ii) Show that the statements $P \leftrightarrow Q$ and $(P \wedge Q) \vee (\sim P \wedge \sim Q)$ are equivalent.